Uncertainty-Guided Agentic Reasoning: Investigating the Impact of Uncertainty Scores on LLM Agents

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Abstract

Current agentic systems demonstrate outstanding performance among various types of tasks, including code generation and validation. However due to probabilistic nature of models and inconsistency in dateset quality, Large Language Models (LLMs), tend to hallucinate, generating structurally correct, but logically wrong code snippets. Hallucination detections methods vary in their methodology, however one specific subset stands out among others. Present papers demonstrated ability of hidden states to store information about models awareness of hallucinated samples. Using this insights it is possible to detect, but not affect generation of hallucinations. In this paper we plan to investigate performance of hallucination aware agentic loops. Resulting system will not only be able to detect hallucinated samples, but also apply this knowledge for generation of correct code.

1 Introduction

2 Novelty and Contributions

Main contributions of this project include:

- **Hallucination Aware Agents:** AS a result it is planned to get an agent that automatically detect and fix hallucinated code snippets
- **Unified Uncertainty Awareness Score Interface** For high modularity and portability, we plan to introduce unified interface, that will allow to use of different implementations of UAS without complex adapters.

3 State-of-the-Art

Currently, most of papers concentrate on detection, but not mitigation of hallucinations

- **Internal Representations as Markers** - various paper proposed approach that allows use internal states as markers for detection of models uncertainty regarding generation of next token. SUch approaches as SPALMA [1], HoloScope [2] and Uncertainty Quantification Heads [3], have different implementations of the same idea - hidden states of model store information about possible hallucination.
- **Middle layers are the most informative** - Many researches [1], [4] demonstrated that most useful information usually stored inside middle layers of models
- **Hallucination detection in code generation** - Some papers [5] proposed idea to use such uncertainty scores for generating reliable code without hallucination

Summing up, although we can pretty efficiently detect hallucination, till this moment there is no proper way to use this information in real, fully autonomous agentic applications

4 Goal of the Thesis

The main goal of this research is to create hallucination aware system, that will use information, extracted from internal states, to detect and fix hallucinations

Specifically, the research aims to:

- Define efficient and clear metrics for hallucination detection
- Develop an agentic loop, that will use defined metrics.
- Evaluate the proposed approach using appropriate datasets and metrics.
- Compare the results with existing methods and analyze the limitations.

5 Detailed Work Plan and Schedule

The whole project planned to span two semesters. First one will be dedicated for exploration of current metrics and wys to detect hallucinations. Additionally it is planned to create first iteration of simple loop, using small local LLM. The second semester will concentrate on improving results obtained in first one

Semester 1

- **Literature review and problem formulation (September–October):**
  - Review relevant scientific literature.
  - Identify limitations of existing approaches.
  - Define evaluation metrics and experimental setup.
- **Methodology and preliminary experiments (November–December):**
  - Select appropriate uncertainty metrics and models.
  - Conduct preliminary experiments.
  - Analyze the initial results.

Semester 2

- **Main experiments and evaluation (January–March):**
  - Implement the proposed methodology.
  - Conduct experiments and compare results.
  - Perform error analysis.
- **Validation and thesis writing (March–April):**
  - Validate the experimental findings.
  - Discuss the results and limitations.
- **Finalization and defense preparation (May):**
  - Finalize the thesis.
  - Prepare the presentation and demonstration.

6 Conclusion

Summing all up, although current approach demonstrated high performance in detecting hallucinations, still there is no reliable and efficient way to use this findings in real world tasks.

This paper is planned to present a solution for this problem by creating unified interface for various hidden states based quality metrics, and fully autonomous agentic loop with hallucination feedback

References

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